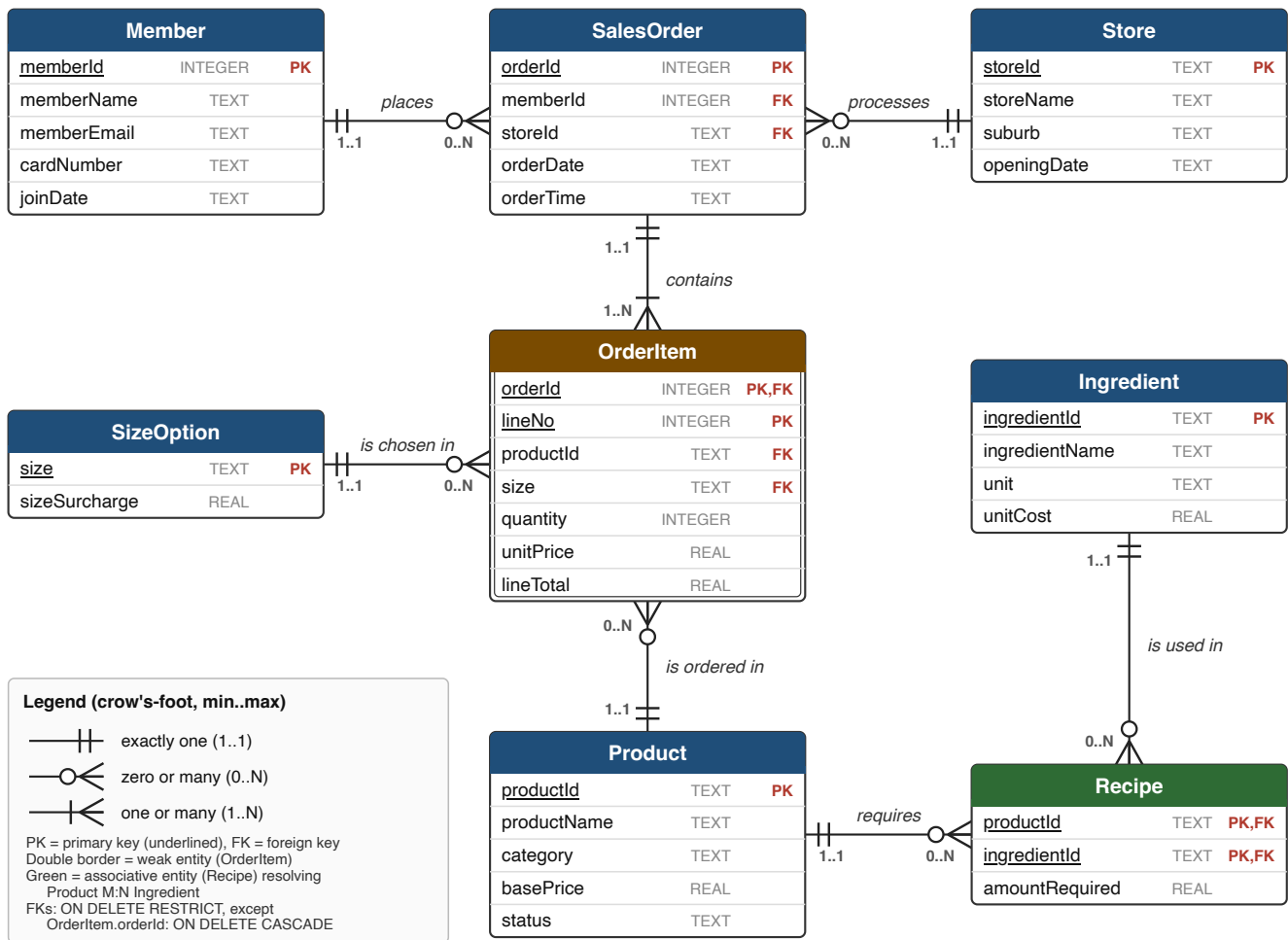


Bubble Trouble – Entity–Relationship Diagram

CITS1402 Project, Semester 2 2026 · Mission 1 · matches createTables.sql



Design commentary

1. Why is OrderItem separate from SalesOrder?

A SalesOrder is one transaction: one member, one store, one date and time. That transaction can contain several different drinks, each with its own product, size, quantity and price. Keeping drinks inside SalesOrder would need repeating columns (drink1, drink2, ...) or would repeat the member, store, date and time on every drink row, causing redundancy and update anomalies. Splitting them gives a 1 : N relationship – one SalesOrder row, one or more OrderItem rows – and lets deleting an order cascade cleanly to its lines.

2. Why is the key of OrderItem naturally composite?

lineNo only numbers the lines *within* one order, so line 1 exists in order 1057 and in order 1058. Neither orderId (many lines per order) nor lineNo (repeats across orders) is unique on its own, but the pair (orderId, lineNo) is. OrderItem is a weak entity: its identity depends on its owning SalesOrder, so the parent's key forms part of its key.

3. Why is Recipe required instead of storing ingredients in Product?

Product–Ingredient is many-to-many: one drink uses many ingredients and one ingredient (e.g. Fresh Milk) is used by many drinks. A list such as “Black Tea, Milk, Pearls” inside Product breaks first normal form, cannot be checked by a foreign key and has nowhere to store how much of each ingredient is needed. Recipe is the associative entity that resolves the M:N into two 1:N relationships: one row per (productId, ingredientId) holding amountRequired, a fact that belongs to the pair rather than to either entity. Its composite key stops an ingredient being listed twice for the same product.

4. Why store the historical unitPrice in OrderItem?

`basePrice + sizeSurcharge` only gives *today's* price. If prices were recalculated after a menu change, yesterday's sales would silently change value, so revenue reports would no longer match what customers actually paid. `unitPrice` is a snapshot of a fact that was true at the moment of sale – deliberate, controlled redundancy rather than an anomaly – and it lets authorised imported sales keep their original price. `lineTotal` likewise records the amount actually charged after any Bubble Rewards free drink.